scalability, and flexibility of letting someone else build, maintain, and manage essential services.

SaaS applications are, necessarily, multi-tenanted, meaning many customers are using the same instance of the service. Multi-tenancy requires that each customer's data be visible to only that customer. But it's even more complicated than that because each business customer will have multiple employees whose access to the system will have to be controlled.

SaaS vendors can't provide the needed levels of functionality and security without strong, flexible authorization systems. Using a policy-based authorization system lets them externalize the logic of access control from the functional aspects of the system. In some cases, like AWS, the policies are visible to the customers, but in most cases, the policies are generated from the SaaS system's UI. Regardless, the use of dynamic authorization allows SaaS vendors to provide the access control needed for their product features and security guarantees while maintaining the flexibility and scalability their services need to compete.

1.3.2 Zero trust security models

Traditionally, companies secured their computer systems using a firewall. The firewall, like a castle wall, is meant to keep attackers out. The firewall carefully controls access to the network and computers inside to ensure only authorized people and systems are granted access.

But perimeter-based security architectures like firewalls are becoming less and less effective. First, mobile devices and remote work mean employees frequently work outside the firewall, necessitating the administrative overhead of virtual private networks (VPNs). Second, the rising use of SaaS-based services means that employees who are inside the firewall are often using systems outside of it. Third, access control needs to be more fine-grained than the coarse-grained "inside or outside the wall" protection that a firewall provides.

As a result, perimeter-based security is being augmented with zero-trust security models. Rather than assuming everyone inside the perimeter is trusted, zero-trust models assume breach and authorize every access.